INVERTIS UNIVERSITY BAREILLY

CENTER FOR SKILL AND ENTREPRENEURSHIP DEVELOPMENT (CSED)

INDUSTRIAL INTERNET OF THINGS

PROGRAM CODE: **IIOT-4**

PROGRAM NAME: **MACHINE LEARNING**

PROJECT NAME: **Healthcare Prediction and Diagnostic Systems**

MENTOR NAME: **MR. DIVYANK CHAUHAN**

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

COURSE NAME: **B.TECH(CSE)**

Section -B

TEAM MEMBERS:	STUDENT ID:
Gaurav Dwivedi(T.L.)	Bcs2024071
Aanya Rastogi	BCS2024072
Aayush Yadav	BCS2024073
Ziya Fatima	BCS2024074
Kashifa Fatima	BCS2024075
Rahat Pathak	BCS2024076
Vikas kumar	BCS2024077
Satyam	BCS2024078
Akhilesh Bhargav	BCS2024081
Vanshpreet Singh	BCS2024083

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Introduction:

Healthcare Prediction and Diagnostic Systems

Healthcare prediction and diagnostic systems are essential components of modern medical and clinical decision-making. In the healthcare sector—ranging from hospitals and diagnostic centers to telemedicine platforms—it is crucial to accurately predict diseases and assist in timely diagnosis to improve patient outcomes.

If diseases are not detected early, they can progress to severe stages, leading to higher treatment costs, increased mortality risk, and strain on healthcare resources. On the other hand, accurate and early prediction enables preventive care, reduces hospitalization rates, and enhances the overall quality of patient care. Therefore, efficient diagnostic systems play a vital role in saving lives and optimizing healthcare services.

Traditionally, diagnosis relied heavily on doctors' experience, manual analysis of reports, and standard clinical tests. While effective, these methods can sometimes be time-consuming and may not capture complex patterns in large volumes of patient data. With the rise of digital health records and advancements in computational power, machine learning techniques have become powerful tools for improving diagnostic accuracy and speed.

This project focuses on using two fundamental machine learning algorithms:

- **Linear Regression**: Used to predict continuous health indicators such as blood pressure, glucose levels, or disease progression trends.
- **Logistic Regression**: Used for classification tasks, such as predicting whether a patient is likely to have a particular disease (e.g., diabetes, heart disease).

By combining prediction and diagnostic decision-making, this system provides a comprehensive solution for healthcare analysis. The integration of statistical models with medical insights enables healthcare providers to make data-driven decisions, reduce diagnostic errors, and improve patient outcomes.

Problem Statement:

In modern healthcare systems, early and accurate disease detection remains a significant challenge due to the increasing volume and complexity of patient data. Traditional diagnostic methods rely heavily on manual analysis, clinical expertise, and standard medical tests, which can be time-consuming and may not always capture hidden patterns or correlations within large datasets.

Diseases such as diabetes and heart conditions often develop gradually and may go undetected until they reach critical stages. Late diagnosis leads to increased treatment costs, higher health risks, and reduced chances of successful recovery. Additionally, healthcare providers face difficulties in identifying high-risk patient groups and delivering personalized treatment due to the lack of data-driven insights.

Therefore, there is a need for an intelligent system that can:

- Accurately predict the likelihood of diseases such as diabetes and heart disease
- Assist in early diagnosis using patient health data
- Group patients based on similar medical characteristics for better treatment planning
- Support healthcare professionals in making faster and more informed decisions

This project aims to develop a machine learning-based healthcare prediction and diagnostic system that leverages techniques such as regression and clustering to enhance diagnostic accuracy, enable early intervention, and improve overall patient care

Objective of the Project:

The primary objectives of this healthcare prediction and diagnostic system are:

- To develop a **machine learning-based system** for early detection of diseases such as diabetes and heart disease
- To improve **diagnostic accuracy** using data-driven approaches instead of relying only on manual methods
- To predict **continuous health indicators** (e.g., glucose level, BP trends) using regression techniques
- To classify patients into **disease / non-disease categories** using classification models
- To implement **patient clustering** for identifying similar health profiles and risk groups
- To assist healthcare professionals in **decision-making and treatment planning**
- To reduce **diagnostic time, cost, and human error**
- To enhance **personalized healthcare and preventive treatment strategies**

Methodology:

The methodology of the project follows a systematic and structured approach, consisting of multiple stages:

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1. Data Collection

- Collect healthcare datasets (Diabetes, Heart Disease)
- Data sources: Kaggle / medical datasets
- Features include: age, glucose, BP, cholesterol, BMI, etc.

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2. Data Preprocessing

- Handle missing values
- Remove noise and outliers
- Normalize / scale data
- Encode categorical variables

---

3. Exploratory Data Analysis (EDA)

- Analyze patterns and relationships
- Use graphs (histograms, heatmaps)
- Identify important features affecting diseases

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4. Feature Selection

- Select relevant attributes
- Remove redundant or less important features
- Improve model efficiency and accuracy

---

5. Model Building

Disease Prediction

- **Linear Regression** → Predict continuous values (e.g., glucose trend)
- **Logistic Regression** → Predict disease (Yes/No)

Clustering

- **K-Means Clustering** → Group patients into:
  - High-risk group
  - Medium-risk group
  - Low-risk group

---

6. Model Training & Testing

- Split dataset into training and testing sets
- Train models using training data
- Test accuracy using test data

---

7. Model Evaluation

- Accuracy
- Precision & Recall
- Confusion Matrix
- F1 Score

---

8. Prediction & Output

- Input patient data
- System predicts:
  - Disease risk (diabetes/heart)
  - Health indicators
  - Patient cluster group

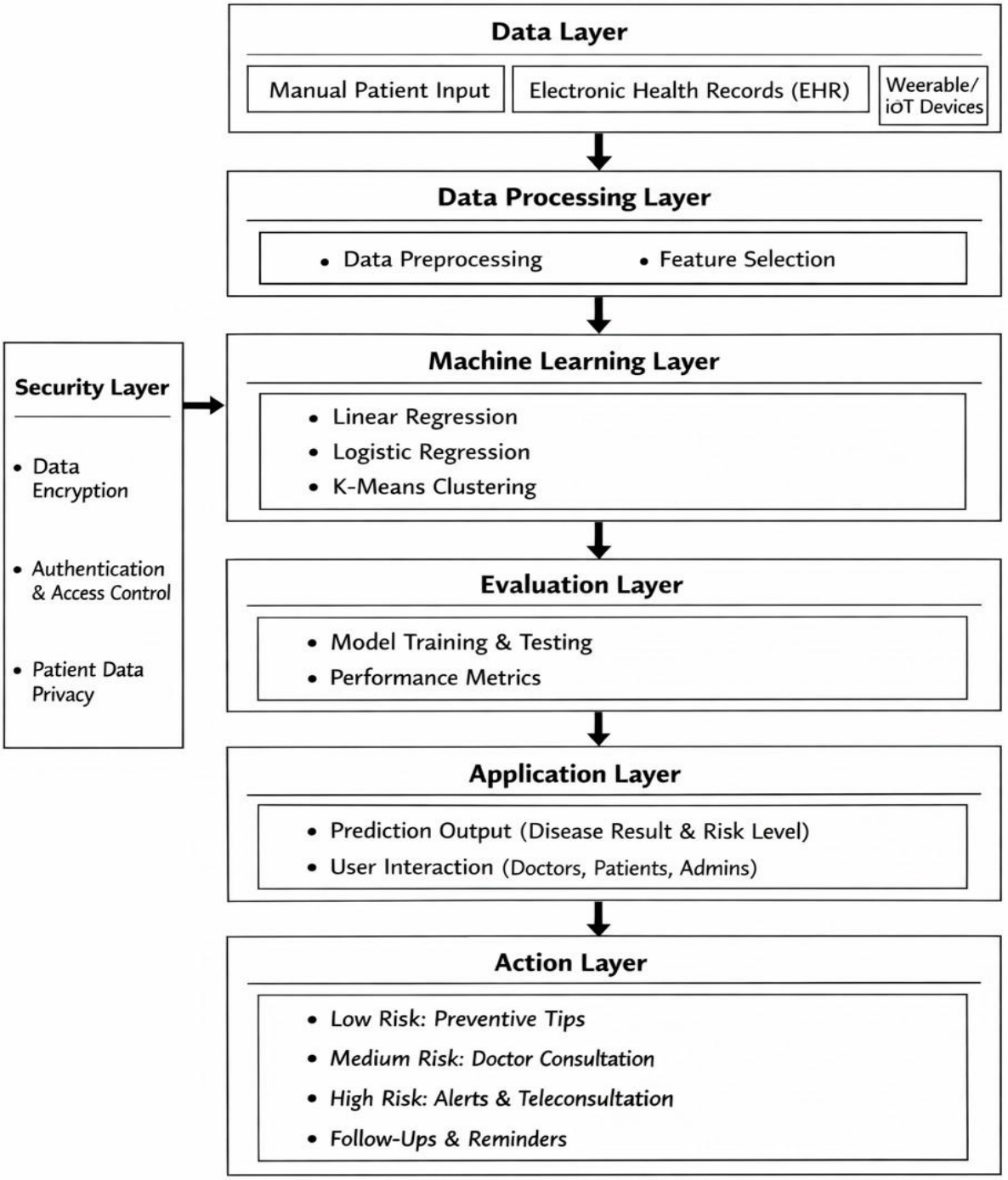
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9. Deployment

- Use **Flask / Streamlit** for web app
- User enters patient data → gets prediction

System Architecture Diagram

Healthcare Prediction and Diagnostic System using Machine Learning



Literature Review:

Healthcare prediction and diagnostic systems have been widely studied with the aim of improving early disease detection and clinical decision-making. Researchers have explored various machine learning techniques such as regression, classification, and clustering to analyze medical data and predict diseases like diabetes and heart disease. Logistic Regression has been extensively used for classification tasks due to its simplicity and effectiveness in binary outcomes, while Linear Regression helps in predicting continuous health parameters. Additionally, clustering techniques such as K-Means have been applied to group patients based on similar health conditions, enabling personalized treatment. Existing studies demonstrate that integrating multiple machine learning approaches enhances prediction accuracy and provides a more comprehensive healthcare solution. However, challenges such as data quality, model interpretability, and real-world implementation still persist, motivating the development of improved systems.

- Machine learning techniques significantly improve **disease prediction accuracy** compared to traditional methods
- **Logistic Regression** is widely used for binary classification problems such as diabetes and heart disease prediction
- **Linear Regression** is effective for predicting continuous medical parameters (e.g., glucose level, BP)
- **K-Means Clustering** helps in grouping patients into risk categories for better analysis
- Combining multiple models leads to **better performance and insights** than single-model systems

References-

Detrano, R. *et al.*
Heart disease prediction using probability algorithms
Chaurasia, V. and Pal, S.
Early prediction of heart diseases using data mining technique

Explanation of Project (Working):

The working of the project can be explained step-by-step as follows:

Initially, the system takes historical sales data as input. This data is then preprocessed to remove inconsistencies and convert it into a usable format. After preprocessing, feature engineering is performed to extract meaningful information from the data.

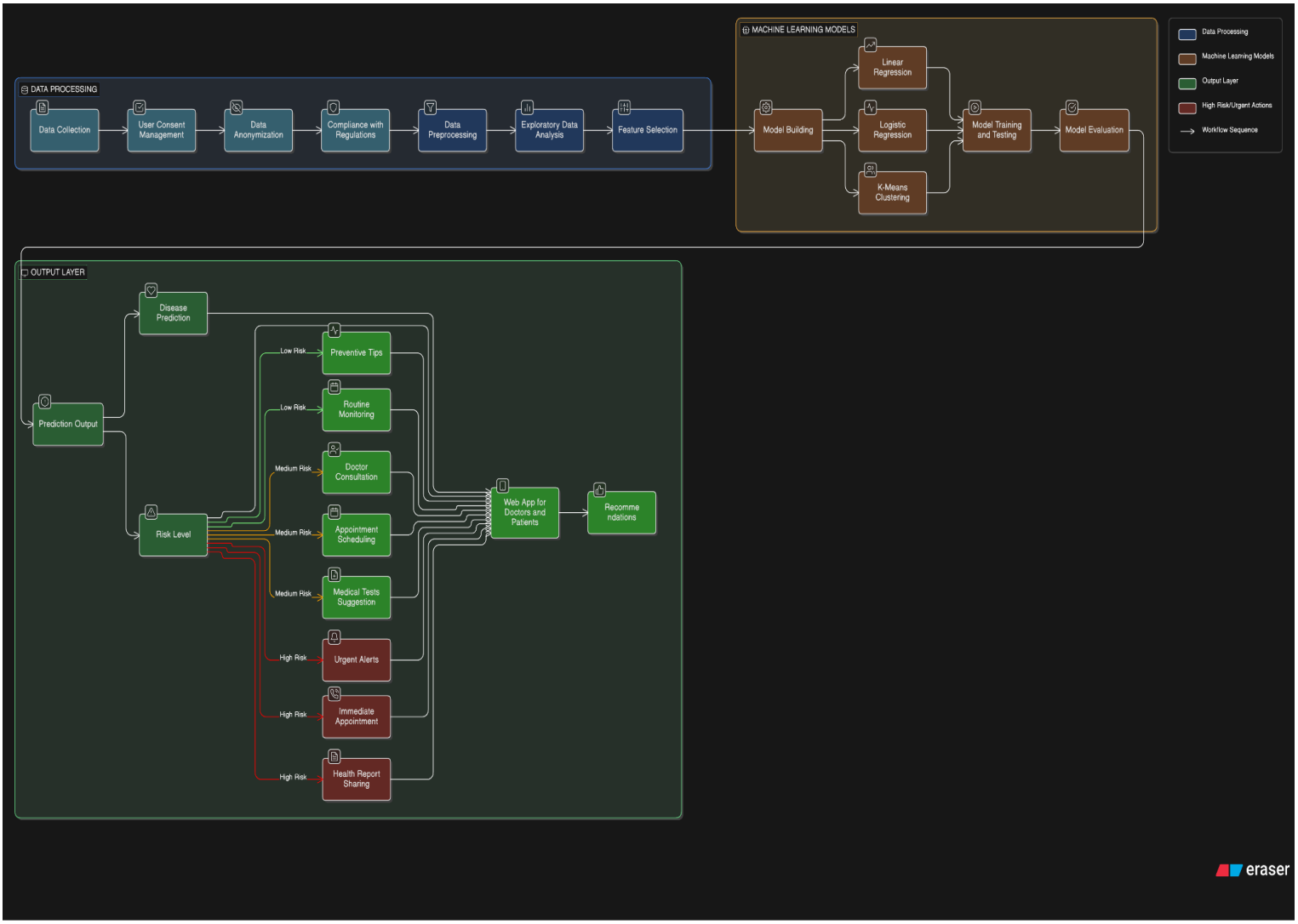
Next, the Linear Regression model is trained using the processed data. This model learns the relationship between time and sales, allowing it to predict future demand values. These predictions help in understanding how much stock will be required in the future.

After predicting demand, the Logistic Regression model is used to make decisions regarding inventory. It takes inputs such as predicted demand and current stock levels and outputs whether the stock should be reordered or not.

Once the decision is made, inventory optimization techniques are applied. EOQ determines the optimal quantity to order, Safety Stock ensures buffer inventory, and Reorder Point identifies the right time to reorder.

Finally, the system displays the results, including predicted demand, reorder decisions, and optimized inventory levels.

Work Flow :



Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset
df = pd.read_csv("heart.csv")

# Show basic info
print(df.head())
print(df.info())

# ----------------------------
# 📊 EDA (Graphs)
# ----------------------------

# 1. Target distribution
sns.countplot(x='target', data=df)
plt.title("Heart Disease Distribution (0 = No, 1 = Yes)")
plt.show()

# 2. Correlation heatmap
plt.figure(figsize=(12,8))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# 3. Age vs Heart Disease
sns.histplot(data=df, x='age', hue='target', bins=30, kde=True)
plt.title("Age Distribution vs Heart Disease")
plt.show()

# 4. Chest Pain Type vs Target
sns.countplot(x='cp', hue='target', data=df)
```

```
plt.title("Chest Pain Type vs Heart Disease")
plt.show()

# ----------------------------
# 🧹 Data
Preprocessing # -----------
-----------------

X = df.drop("target", axis=1)
y = df["target"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# ----------------------------
# 🤖 Models
# ----------------------------

models = {
    "Logistic Regression": LogisticRegression(),
    "Random Forest": RandomForestClassifier(),
    "KNN": KNeighborsClassifier()
}

results = {}

for name, model in models.items():
```

```
model.fit(X_train, y_train)
```

```
    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    results[name] = acc

    print(f"\n{name} Results:")
    print("Accuracy:", acc)
    print(confusion_matrix(y_test, y_pred))
    print(classification_report(y_test, y_pred))

# ----------------------------
# 📊 Accuracy Comparison
Graph # ----------------------------

plt.bar(results.keys(), results.values())
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.show()

# ----------------------------
# 🌟 Feature Importance (Random
Forest) # ----------------------------

rf = RandomForestClassifier()
rf.fit(X_train, y_train)

importance = rf.feature_importances_
features = df.columns[:-1]

plt.figure(figsize=(10,6))
sns.barplot(x=importance, y=features)
plt.title("Feature Importance")
plt.show()
```

1. Heart Disease:

Introduction:-

Heart disease is one of the leading causes of death globally and poses a serious threat to human health. It includes conditions such as coronary artery disease, heart attack, and other cardiovascular disorders that affect the normal functioning of the heart. Major risk factors include high blood pressure, high cholesterol, diabetes, smoking, obesity, and lack of physical activity.

With the advancement of technology, Artificial Intelligence (AI) and Machine Learning (ML) have become powerful tools in the healthcare sector. These technologies help in analyzing large amounts of medical data to identify patterns and predict diseases at an early stage.

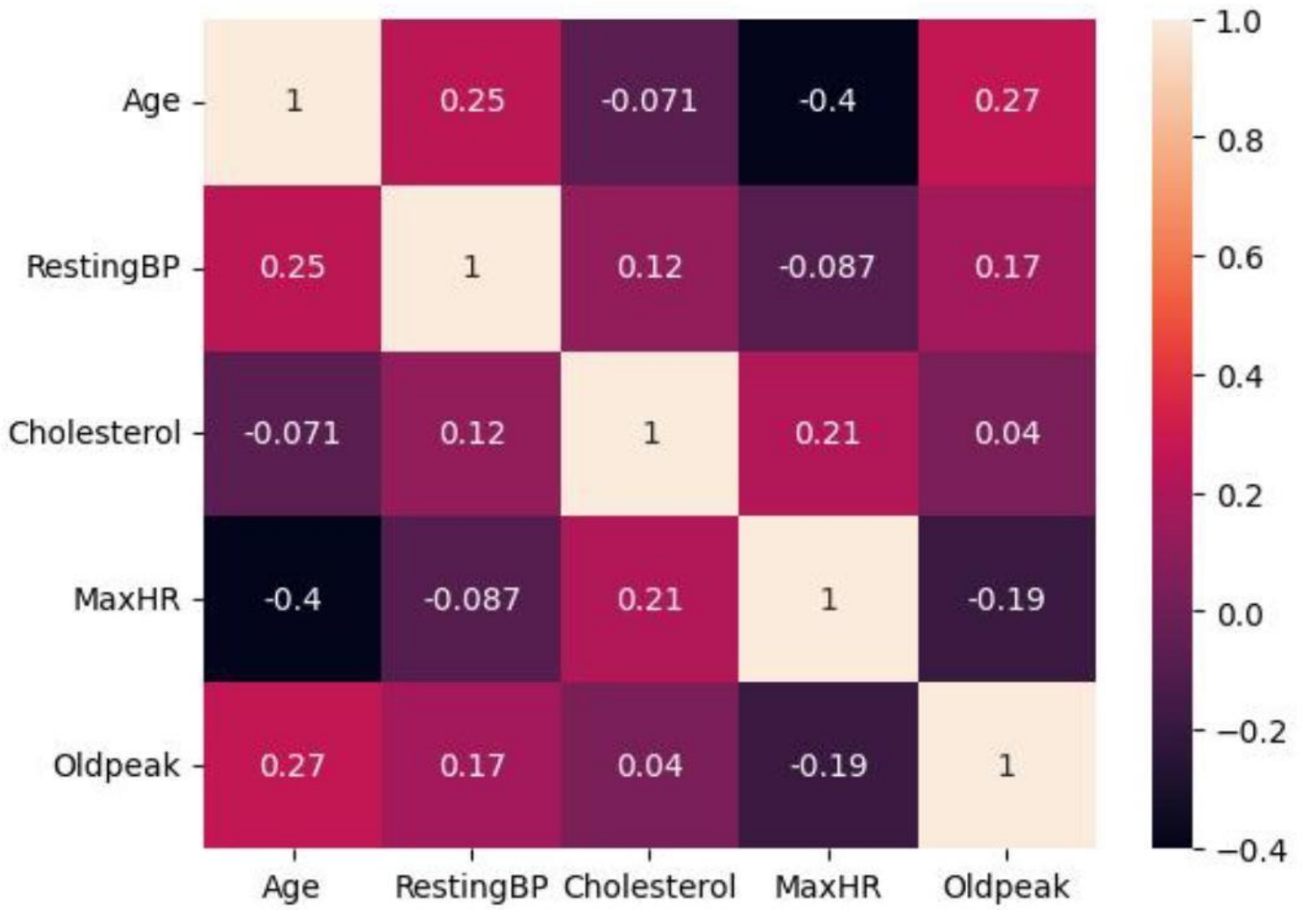
In this project, the Heart Disease Prediction System is designed to predict whether a person is likely to have heart disease based on various medical attributes such as age, gender, blood pressure, cholesterol level, and heart rate. The system uses machine learning algorithms to process the data and provide accurate predictions.

The main objective of this module is to assist doctors and healthcare professionals in making faster and more accurate decisions. Early prediction of heart disease can help in timely treatment and prevention, thereby reducing the risk of severe health complications.

Overall, this system aims to improve healthcare services by combining medical knowledge with intelligent data-driven techniques.

Graphs & Descriptions:-

1. Correlation Heatmap of Heart Disease Features -



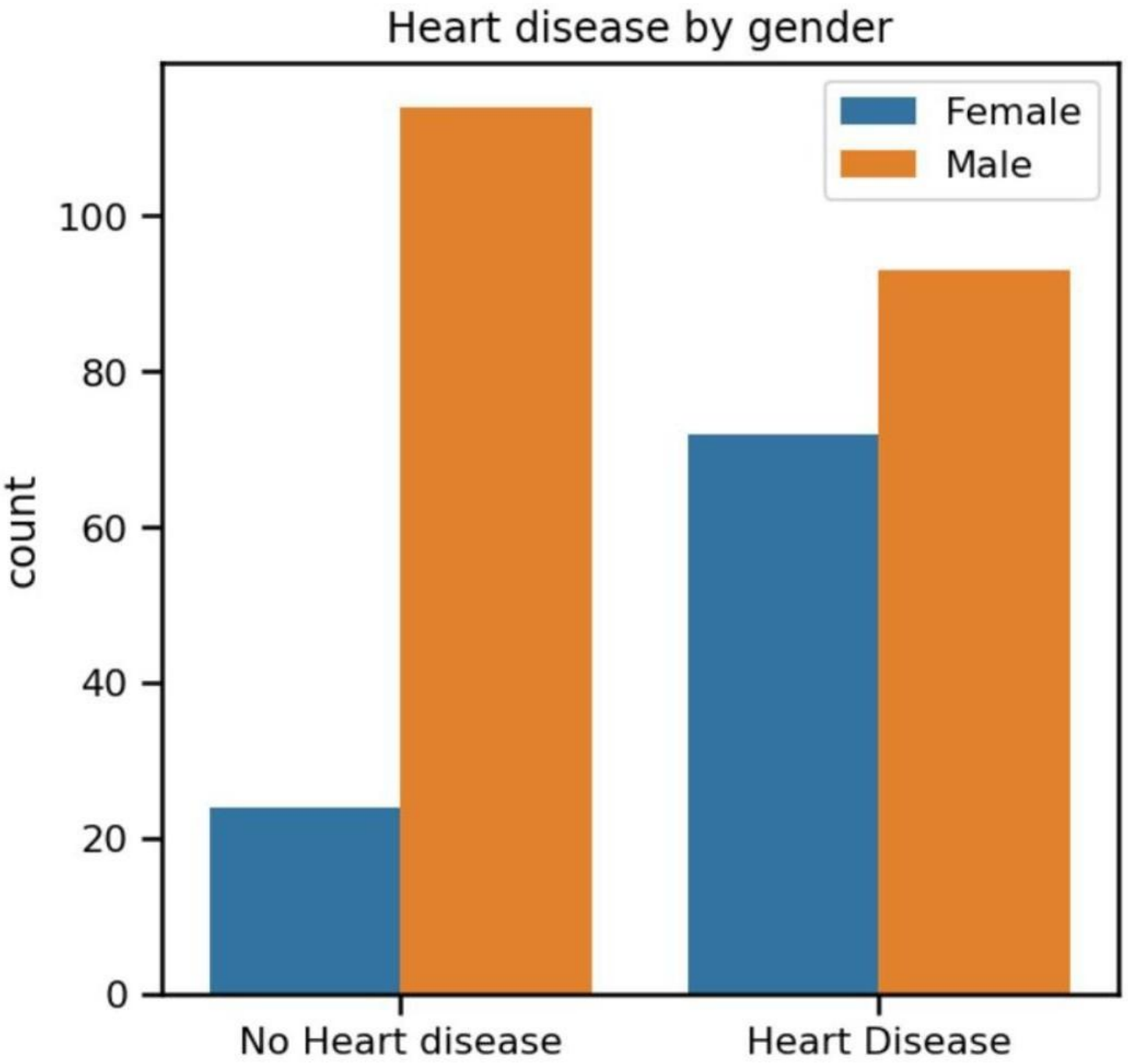
Description:

This graph is a **correlation heatmap** that shows the relationship between different medical features such as Age, Resting Blood Pressure (RestingBP), Cholesterol, Maximum Heart Rate (MaxHR), and Oldpeak.

- Values range from **-1 to +1**
  - **+1** ⟶ strong positive correlation
  - **-1** ⟶ strong negative correlation
- Age and MaxHR show a **negative correlation (-0.4)**, meaning as age increases, maximum heart rate tends to decrease.
- Cholesterol has a **weak correlation** with other features, indicating it may not strongly influence other variables independently.
- Oldpeak shows a **slight positive relationship with Age (0.27)**.

This heatmap helps in **feature selection** and understanding how variables influence each other in predicting heart disease.

2. Heart Disease Distribution by Gender -



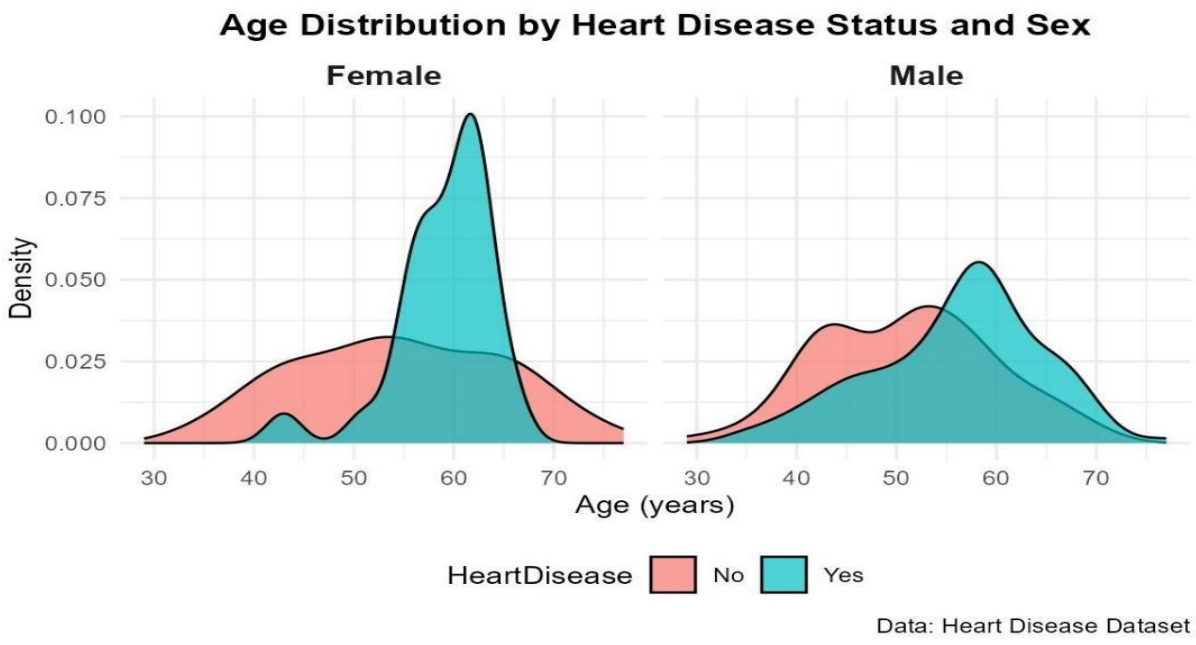
Description:

This graph is a **count plot (bar chart)** showing the number of males and females with and without heart disease.

- Males have a **higher number of heart disease cases** compared to females.
- The number of males without heart disease is also higher, indicating a **larger male dataset representation**.
- Females show comparatively **lower counts in both categories**.

This graph highlights that **gender plays a significant role** and should be considered as an important feature in prediction models.

3: Age Distribution by Heart Disease Status and Gender -



Description:

This graph is a **KDE (Kernel Density Estimation) plot**, showing the distribution of age based on heart disease status for both males and females.

- Most heart disease cases occur in the age group of **50–65 years**.
- In females, the peak density for heart disease is around **55–60 years**.
- In males, heart disease is more spread out but still peaks around **50–60 years**.
- Younger individuals show **lower probability** of heart disease.

This graph indicates that **age is a major risk factor** in heart disease prediction.

2. Diabetes Disease:

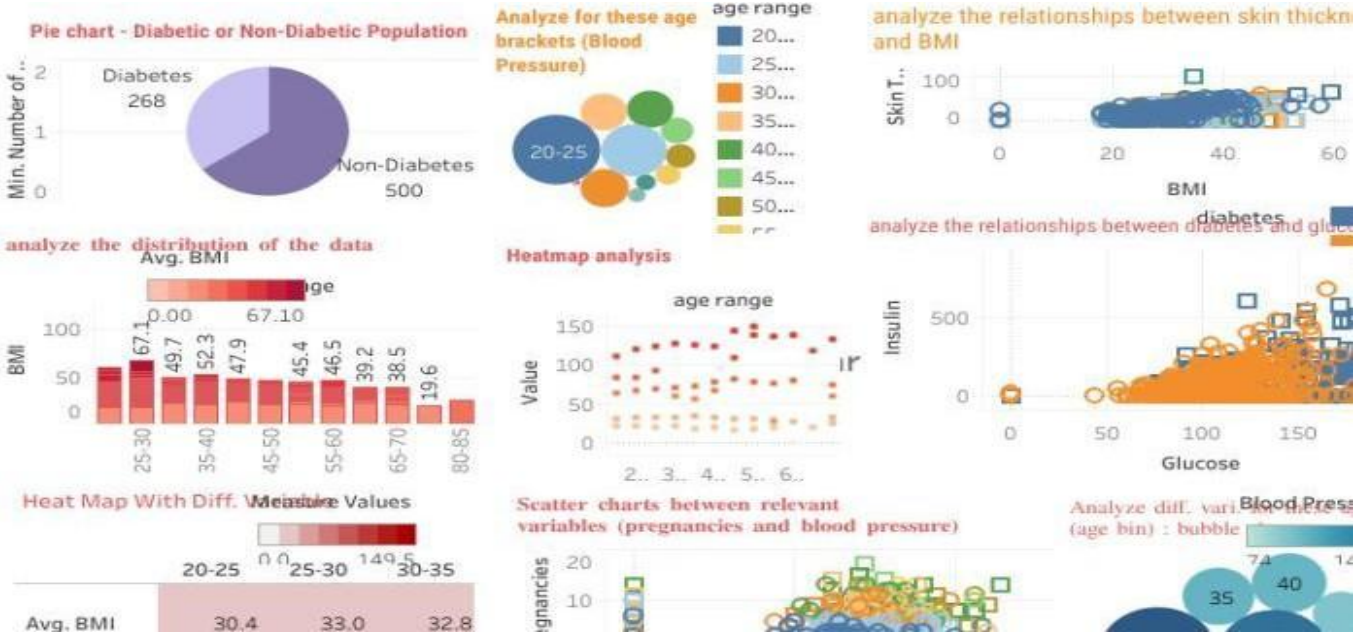
Introduction:-

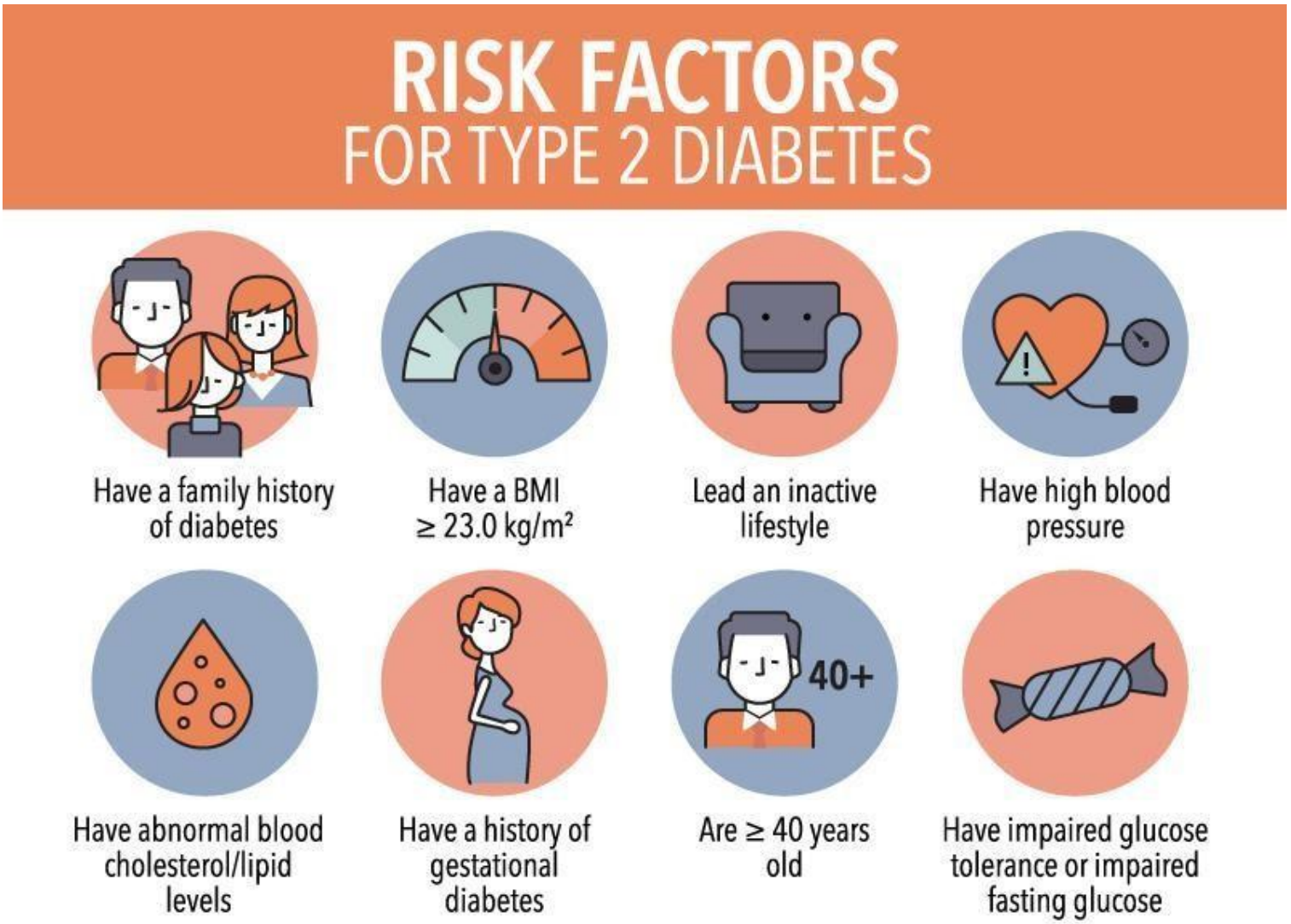
Diabetes is a chronic disease that occurs when the body cannot properly control blood sugar levels due to lack of insulin or ineffective use of insulin. It is one of the most common health problems worldwide and can lead to serious complications such as heart disease, kidney damage, and vision loss if not managed properly.

In this project, the Diabetes Prediction Module uses Artificial Intelligence (AI) and Machine Learning (ML) to analyze patient data such as glucose level, blood pressure, BMI, insulin level, and age. Based on these parameters, the system predicts the likelihood of diabetes.

The main goal of this module is to enable early detection and support healthcare professionals in making accurate decisions. Early prediction helps in timely treatment and encourages individuals to adopt healthier lifestyles, reducing the risk of severe complications.

Diabetes Disease Prediction (Overview)





Diabetes prediction uses **Machine Learning (ML)** to predict whether a person is diabetic based on medical data like:

- Glucose level
- BMI
- Age
- Blood pressure
- Insulin level

ML Models Used

Common algorithms:

- Logistic Regression (best for binary classification)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

Dataset

Most projects use:
Pima Indians Diabetes Dataset

Features:

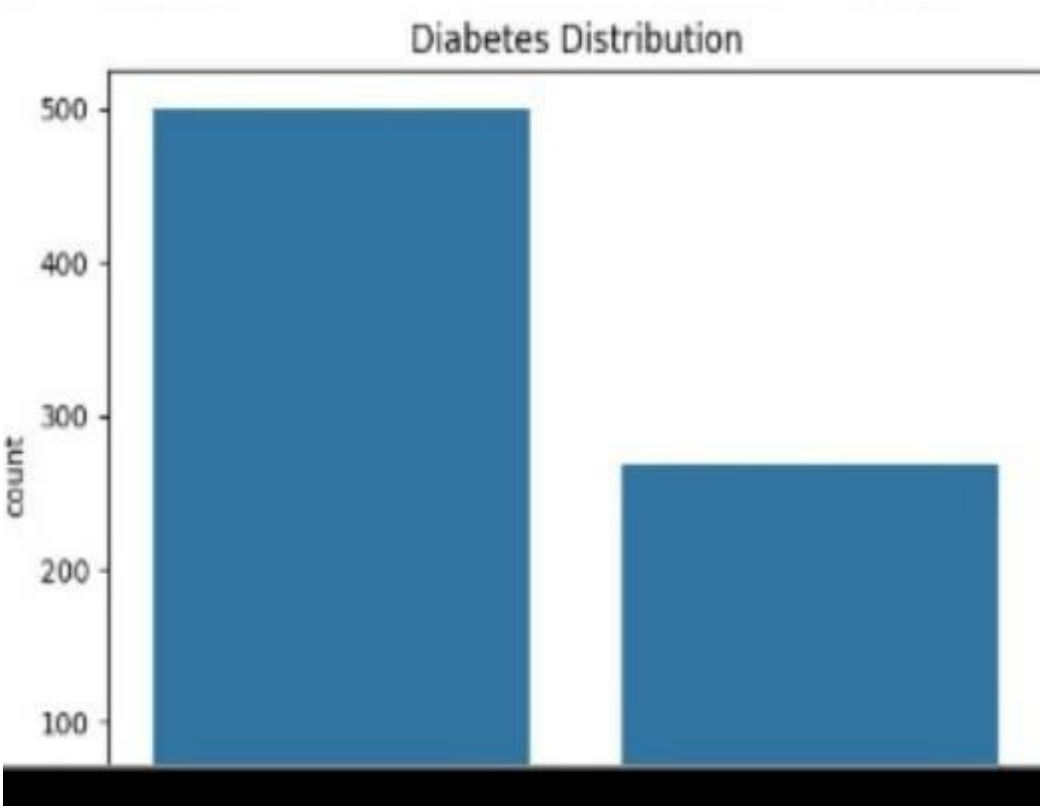
- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

Target:

- `0 = Non-Diabetic`
- `1 = Diabetic`

Graphs & Descriptions:-

1. Diabetes Distribution -



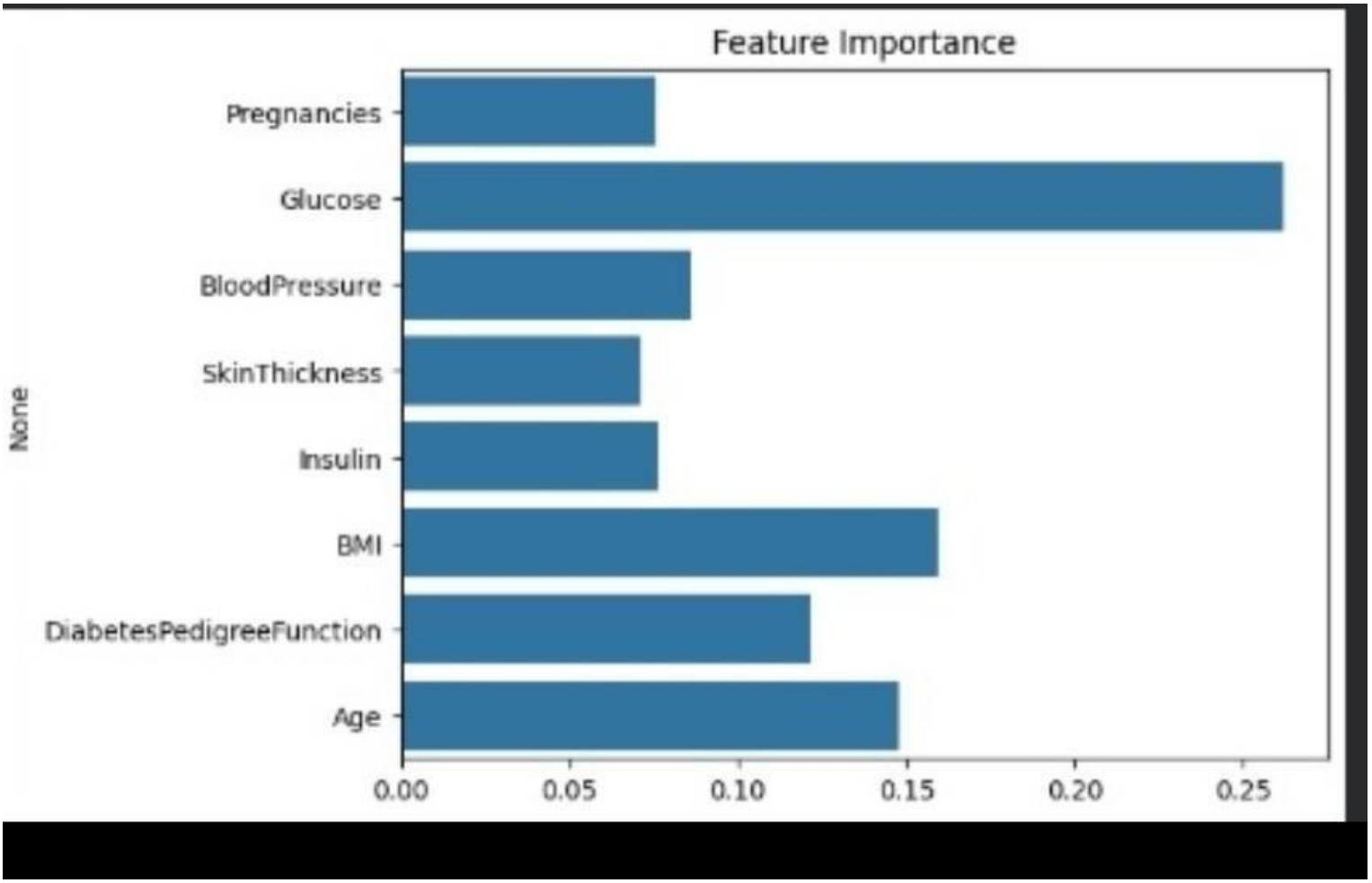
Description:

This graph represents the distribution of diabetes cases in the dataset.

- It shows the count of people with and without diabetes.
- The number of non-diabetic cases is higher compared to diabetic cases.
- This indicates a class imbalance in the dataset.

This graph helps in understanding dataset composition and is important for model training and evaluation.

2. Confusion Matrix for Random Forest-



Description:

This graph shows the performance of the Random Forest model.

- 80 (True Negatives) ⟶ correct non-diabetic predictions
- 38 (True Positives) ⟶ correct diabetic predictions
- 19 (False Positives) ⟶ incorrect diabetic predictions
- 17 (False Negatives) ⟶ missed diabetic cases

Compared to Logistic Regression, Random Forest shows slightly better prediction of diabetic cases, making it more effective.

3. Parent Clustering Disease:

Introduction:-

Patient clustering is an important technique in healthcare that uses machine learning to group individuals based on similar medical characteristics. Unlike traditional classification methods, clustering is an unsupervised learning approach that does not require labeled data.

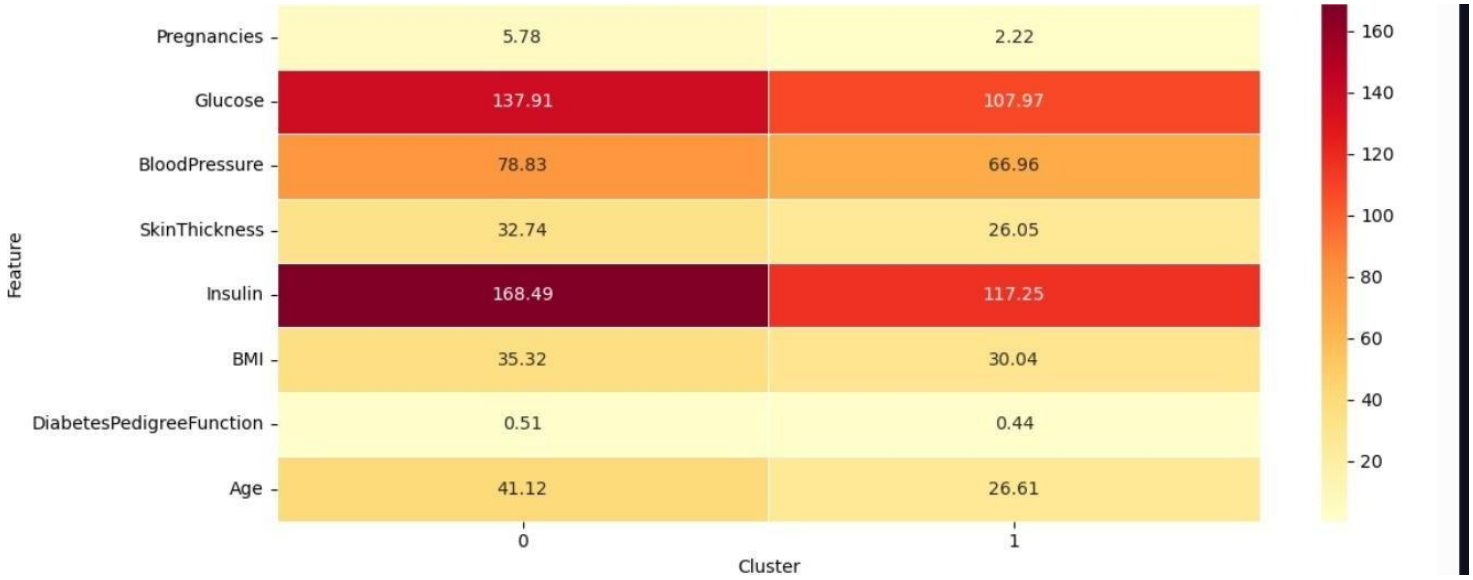
In this project, clustering is applied to medical datasets containing features such as glucose level, blood pressure, BMI, insulin level, age, and other health indicators. Algorithms like K-Means are used to divide patients into different clusters based on similarities in their data.

The main objective of patient clustering is to identify hidden patterns and group patients into categories such as high-risk and low-risk groups. This helps healthcare professionals in better understanding patient conditions, providing personalized treatment, and improving early diagnosis.

By analyzing clusters, it becomes easier to detect patterns that may not be visible through manual analysis, thus enhancing decision-making in healthcare systems.

Graphs & Descriptions:-

1. Cluster-wise Feature Mean Heatmap-



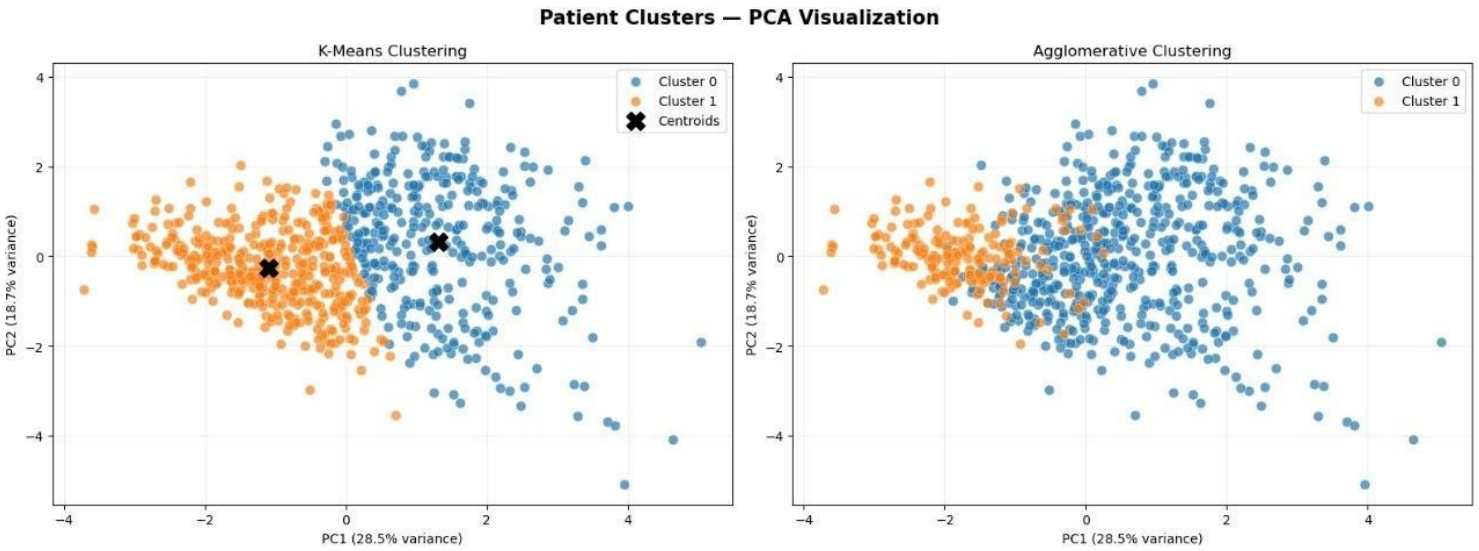
Description:

This graph is a **heatmap showing the average (mean) values of features for each cluster**.

- Each row represents a feature, and columns represent clusters (0 and 1).
- Warmer colors (red/orange) indicate **higher values**, while lighter colors indicate lower values.
- Cluster 0 shows higher values for:
  - Glucose (137.91)
  - Insulin (168.49)
  - BMI (35.32)
  - Age (41.12)
- Cluster 1 has comparatively **lower values**, indicating lower risk.

This heatmap clearly highlights the **difference between clusters**, helping identify which cluster is more prone to disease.

2. Patient Clusters — PCA Visualization -



Description:

Both graphs show patient data projected onto two principal components — PC1 (28.5% variance) and PC2 (18.7% variance) — revealing two distinct patient groups:

- Cluster 0 (Blue): Larger group, spread toward positive PC1
- Cluster 1 (Orange): Denser group, concentrated in negative PC1

In K-Means, black ✕ marks show centroids with clear geometric separation. In Agglomerative Clustering, boundaries are slightly irregular due to its hierarchical linkage approach. Both methods produce similar clustering patterns, confirming two stable and meaningful patient subgroups in the disease dataset.

Result & Discussions :

The results of the project show that the machine learning models are effective in predicting demand and optimizing inventory.

The Linear Regression model successfully captures the trend in sales data and provides reasonably accurate predictions. The Logistic Regression model effectively classifies whether stock needs to be reordered, helping automate decision-making.

The use of inventory optimization techniques further improves efficiency by minimizing costs and ensuring availability of products.

However, the performance of the models depends on the quality and quantity of data. In cases where data is limited or highly non-linear, the accuracy may decrease. Advanced models can be used to improve performance in such scenarios.

Future Scope:

the proposed system can be further enhanced and expanded in several ways:

- **Integration with Real-Time Data**
  Incorporate real-time patient data from wearable devices and IoT-based health monitoring systems
- **Use of Advanced Machine Learning Models**
  Implement advanced algorithms such as Random Forest, Support Vector Machines (SVM), and Deep Learning for improved accuracy
- **Expansion to More Diseases**
  Extend the system to predict additional diseases such as cancer, kidney disorders, and neurological conditions
- **Development of a Web/Mobile Application**
  Create a user-friendly interface for doctors and patients using technologies like Flask, Django, or mobile apps
- **Incorporation of Electronic Health Records (EHR)**
  Integrate with hospital databases for large-scale and real-world deployment
- **Explainable AI (XAI)**
  Add interpretability features to explain predictions, increasing trust among healthcare professionals
- **Cloud-Based Deployment**
  Deploy the system on cloud platforms for scalability and remote accessibility
- **Personalized Recommendation System**
  Provide lifestyle and treatment suggestions based on patient health data

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Conclusion:

This project presents a machine learning-based healthcare prediction and diagnostic system aimed at improving early disease detection and patient analysis. By utilizing Linear Regression for predicting continuous health parameters, Logistic Regression for disease classification, and K-Means Clustering for patient grouping, the system provides a comprehensive and integrated approach to healthcare analytics.

The developed model successfully demonstrates how data-driven techniques can assist in identifying diseases such as diabetes and heart conditions at an early stage. Additionally, patient clustering enables the categorization of individuals into different risk groups, supporting personalized treatment planning and efficient resource allocation.

Overall, the system enhances diagnostic accuracy, reduces dependency on manual analysis, and supports healthcare professionals in making informed decisions. This project highlights the potential of machine learning in transforming traditional healthcare systems into more intelligent, efficient, and patient-centric solutions.

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